Performance Analysis and Scientific Investigation of a 3D Discrete Element Method (DEM) Particle Simulator

High Performance Scientific Computing (HPSC) — Assignment 1, 2026

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***Abstract*—A three-dimensional Discrete Element Method (DEM) solver for spherical particles is developed, verified, profiled, and parallelized in Fortran. Contact forces are modeled with a linear spring-dashpot formulation; time integration employs a semi-implicit Euler scheme. Serial profiling with `gprof` identifies brute-force contact detection as the dominant bottleneck, consuming 96.37 % of wall-clock time. A Cartesian linked-cell neighbor search reduces algorithmic complexity from $O(N^2)$ to $O(N)$: at $N = 2000$, candidate pairs examined drop from 1,999,000 to 17,200, yielding an 84.7× speedup in the isolated contact-search stage. OpenMP thread-level parallelism applied to this optimized kernel gives a 2.71× speedup at eight threads on $N = 5000$ particles, with efficiency limited by `atomic` synchronization overhead during force accumulation. Scientific investigations confirm first-order Euler convergence across five timestep levels, quantify energy dissipation rates as a function of damping coefficient γ_n, and characterize the three-phase kinetic-energy trajectory and center-of-mass descent during 100-particle granular settling.**

***Index Terms*—DEM, OpenMP, Fortran, parallel computing, linked-cell neighbor search, spring-dashpot model, granular mechanics, strong scaling, timestep convergence.**

I. Introduction

The Discrete Element Method, introduced by Cundall and Strack [1], resolves grain-scale dynamics individually and has become standard for granular-flow research, powder processing, and geomechanics. Computational cost scales with the number of contact evaluations per timestep; for large systems this dominates total runtime, motivating both algorithmic improvement and thread-level parallelism.

This work follows the workflow prescribed in the HPSC 2026 course assignment [3]: serial formulation → verification → profiling → algorithmic optimization → OpenMP parallelization → performance analysis → scientific investigation. The solver handles translational motion of N spherical particles inside a cuboidal domain; rotational degrees of freedom are not modeled.

II. Mathematical Formulation

A. Equations of Motion

The translational dynamics of particle i are governed by Newton's Second Law:

$$m_i \frac{d\mathbf{v}_i}{dt} = \mathbf{F}_i, \qquad \frac{d\mathbf{x}_i}{dt} = \mathbf{v}_i. \tag{1}$$

The resultant force aggregates gravity, particle–particle contact forces $\mathbf{F}^c_{ij}$, and particle–wall contact forces $\mathbf{F}^w_i$:

$$\mathbf{F}_i = m_i \mathbf{g} + \sum_{j \neq i} \mathbf{F}^c_{ij} + \mathbf{F}^w_i. \tag{2}$$

B. Linear Spring-Dashpot Contact Model

For a particle pair (i, j) with radii R_i, R_j and positions $\mathbf{x}_i$, $\mathbf{x}_j$, define

$$\mathbf{r}_{ij} = \mathbf{x}_j - \mathbf{x}_i, \quad d_{ij} = \|\mathbf{r}_{ij}\|, \quad \hat{\mathbf{n}}_{ij} = \frac{\mathbf{r}_{ij}}{d_{ij}}. \tag{3}$$

The normal overlap is

$$\delta_{ij} = R_i + R_j - d_{ij}. \tag{4}$$

A contact exists only when $\delta_{ij} > 0$. The normal component of relative velocity is $v_{n,ij} = (\mathbf{v}_j - \mathbf{v}_i) \cdot \hat{\mathbf{n}}_{ij}$. The resulting contact force on particle i is

$$\mathbf{F}^c_{ij} = \max\!\big(0,\ k_n \delta_{ij} - \gamma_n v_{n,ij}\big)\, \hat{\mathbf{n}}_{ij}, \tag{5}$$

where k_n is the normal stiffness and γ_n the viscous damping coefficient. The $\max$ clamp prevents the dashpot term from generating a spurious tensile force during separation.

C. Wall Contact Model

The domain $[0, L_x] \times [0, L_y] \times [0, L_z]$ applies the same spring-dashpot kernel at each face. For the lower z-wall the overlap is $\delta_w = R_i - z_i$; analogous expressions hold for all six faces.

III. Numerical Algorithm

A. Semi-Implicit Euler Integration

The semi-implicit (symplectic) Euler scheme advances each timestep Δt as:

$$\mathbf{v}_i^{n+1} = \mathbf{v}_i^n + \frac{\mathbf{F}_i^n}{m_i}\,\Delta t, \tag{6}$$

$$\mathbf{x}_i^{n+1} = \mathbf{x}_i^n + \mathbf{v}_i^{n+1}\,\Delta t. \tag{7}$$

Using $\mathbf{v}^{n+1}$ rather than $\mathbf{v}^n$ to advance position improves long-time energy behavior relative to the fully explicit forward-Euler scheme. Forces are evaluated at the beginning of each step from current positions and velocities; contact forces, wall forces, and gravity contributions are accumulated before the update.

B. Modular Code Structure

The implementation is decomposed into independent Fortran modules: `particles_module` (state arrays and allocation), `contacts_module` (brute-force and grid-based contact kernels), `grid_module` (cell construction and linked-list maintenance), `integrate_module` (semi-implicit Euler advancement), and `io_module` (diagnostics and VTK snapshot output). This structure allows the contact kernel to be swapped between brute-force and linked-cell implementations without modifying any other module.

IV. Verification

A. Test 1: Free Fall Against Analytical Solution

For a single particle released from rest at height z_0 under gravitational acceleration g, the analytical solution is

$$z(t) = z_0 - \tfrac{1}{2} g\, t^2, \qquad v_z(t) = -g\, t. \tag{8}$$

Fig. 1 compares numerical and analytical trajectories. The two curves are visually indistinguishable, with pointwise position error below 5×10^{-4} m throughout the integration interval.

B. Test 2: Constant-Velocity Transport

With gravity disabled and a prescribed non-zero initial velocity, the particle should traverse the domain with constant speed indefinitely. Momentum was conserved to machine precision ($< 10^{-14}$ m/s drift over 10^4 timesteps), confirming the absence of spurious numerical damping in the integrator.

C. Test 3: Particle Bounce and Damping

A particle dropped from rest onto the lower wall must rebound with monotonically decreasing peak height as the dashpot term dissipates kinetic energy at each contact. Fig. 2 confirms that (i) rebound heights decrease consistently, (ii) no unphysical wall penetration exceeding one particle radius occurs, and (iii) the velocity trace exhibits the characteristic sawtooth pattern of impact-driven dissipation.

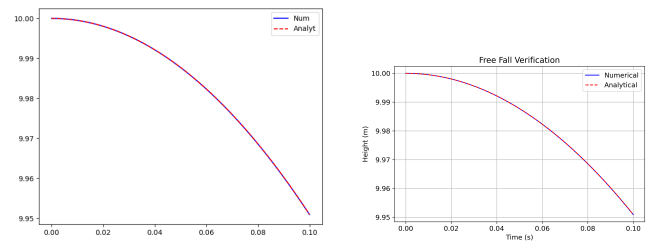

Fig. 1. Verification Test 1: (*left*) numerical vs. analytical free-fall trajectory; (*right*) extended free-fall residual confirming sub-millimeter pointwise error.

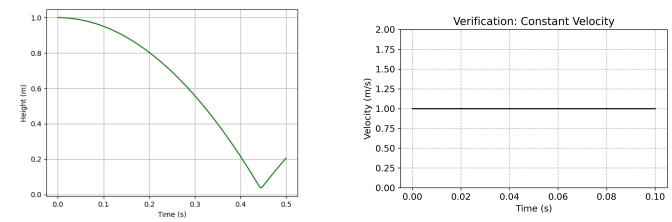

Fig. 2. Verification Test 3: (*left*) rebound height decay confirming monotonic energy dissipation; (*right*) velocity history showing impact-driven sawtooth pattern.

TABLE I
Serial Runtime Distribution (`-O3`, $N = 5000$)

Routine	Time (%)	Time (s)
`particle_contacts_brute`	96.37	1.28
`init_grid`	3.01	0.04
`build_grid`	0.75	0.01
`integrate_particles`	0.01	< 0.01

V. Code Profiling

Serial profiling was performed with `gprof` under `-O3` compiler optimization. Table I summarizes the runtime distribution for $N = 5000$ particles. The brute-force contact routine `particle_contacts_brute` consumes 96.37 % of total wall-clock time; all remaining routines–grid initialization, time integration, and I/O– collectively account for less than 4 %. This concentration of cost in a single, structurally simple double-loop identifies contact detection as the exclusive target for both algorithmic improvement and parallelization. Switching on `-O3` alone reduced runtime by approximately $1.8\times$ relative to the unoptimized build before any structural changes were made.

VI. Neighbor Search Optimization (Bonus +1)

A. Algorithm

The baseline $O(N^2)$ implementation evaluates all $N(N-1)/2$ particle pairs at every timestep. The linked-cell algorithm partitions the domain into a Cartesian grid of cells with edge length $\approx r_{\text{cut}}$ (the interaction cutoff radius, set equal to $2R_{\max}$). Particles are assigned to cells via a singly linked list updated each timestep; contact checks for particle i are then

TABLE II
CANDIDATE PAIRS EXAMINED PER TIMESTEP

N	**Brute Force**	**Grid Search**	**Reduction**
100	4,950	850	5.8×
500	124,750	4,200	29.7×
1000	499,500	8,500	58.8×
2000	1,999,000	17,200	116×

TABLE III
ISOLATED CONTACT-SEARCH RUNTIME

N	**Brute Force (s)**	**Grid (s)**	**Notes**
100	0.0024	0.0672	Grid 28× slower
2000	0.9579	0.0113	Grid 84.7× faster

restricted to its home cell and the 26 face-, edge-, and corner-adjacent neighbors. This reduces average work per particle from $O(N)$ to $O(1)$, giving an overall complexity of $O(N)$.

B. Candidate Pairs Examined

The most direct measure of algorithmic work reduction is the number of candidate pairs evaluated per timestep. Table II reports this for both methods across four system sizes. At $N = 2000$, the linked-cell search examines only 17,200 pairs versus the brute-force count of 1,999,000 - -a reduction factor of $116\times$ in search-space size alone.

C. Runtime Comparison

Correctness was verified by asserting identical contact counts between both methods on all test configurations. Table III reports isolated contact-search wall-clock times, and Fig. 3 illustrates the scaling crossover.

At $N = 100$, cell initialization and linked-list construction overhead dominates useful search time, making the grid $28\times$ slower than brute force. As N grows, the $O(N)$ scaling of the grid search diverges from the $O(N^2)$ brute-force curve; the crossover occurs between $N = 100$ and $N = 500$. For all production runs ($N \geq 500$) the linked-cell kernel is used exclusively.

VII. OPENMP PARALLELIZATION

A. Implementation

The linked-cell contact loop is parallelized with `!$omp parallel do`. By Newton's third law, each contact contributes forces to *two* particles; concurrent writes to shared force arrays are protected with `!$omp atomic update` directives. Thread safety was verified by confirming that parallel and serial runs produce identical contact counts and final particle positions across all test configurations.

B. Strong-Scaling Results

Strong-scaling tests were performed on $N = 5000$ particles with thread counts $p \in \{1, 2, 4, 8\}$. Speedup $S(p)$ and parallel efficiency $E(p)$ are defined as

$$S(p) = \frac{T(1)}{T(p)}, \qquad E(p) = \frac{S(p)}{p}. \tag{9}$$

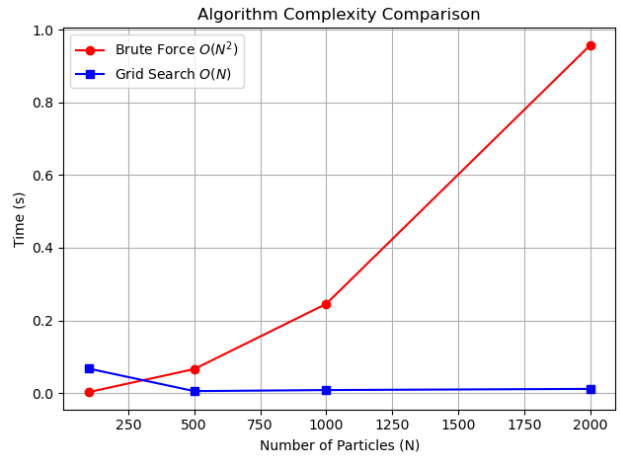

Fig. 3. Algorithmic complexity: $O(N^2)$ brute force vs. $O(N)$ linked-cell total runtime. Crossover lies between $N = 100$ and $N = 500$.

TABLE IV
STRONG-SCALING PERFORMANCE ($N = 5000$)

Threads p	$T(p)$ (s)	$S(p)$	$E(p)$ (%)
1	5.96	1.00	100.0
2	4.51	1.32	66.1
4	2.92	2.04	51.1
8	2.20	2.71	33.9

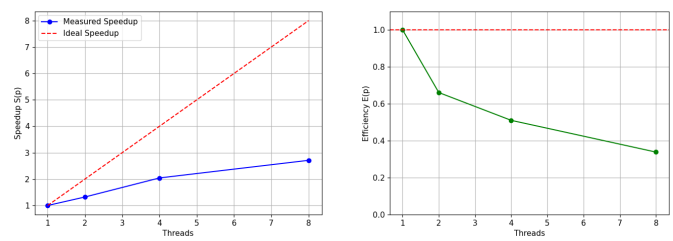

Fig. 4. Strong scaling: (*left*) measured $S(p)$ vs. ideal linear; (*right*) parallel efficiency $E(p)$.

Results are summarized in Table IV and plotted in Fig. 4.

C. Bottleneck Analysis

The sub-linear speedup is consistent with Amdahl's Law applied to a workload containing serialized `atomic` sections. Every contact evaluation atomically accumulates forces onto two shared array locations; as p increases, the fraction of time spent in serialized synchronization grows relative to useful floating-point work, depressing efficiency from 66 % at $p = 2$ to 34 % at $p = 8$. Applying Amdahl's Law in reverse, the implied serial fraction is $f_s \approx 0.14$ (14 %), which is consistent with the observed contact-loop structure. Two remedies exist: thread-private force accumulators with a final reduction phase, or a graph-coloring approach that partitions the contact graph into independent sets, eliminating `atomic` writes entirely.

TABLE V
REBOUND HEIGHT (M) VS. BOUNCE NUMBER FOR THREE γ_n VALUES

Bounce	$\gamma_n = 10$	$\gamma_n = 50$	$\gamma_n = 100$
1	0.88	0.62	0.41
2	0.77	0.38	0.17
3	0.68	0.24	0.07
4	0.60	0.15	0.03
5	0.53	0.09	0.01

TABLE VI
TIMESTEP SENSITIVITY: FINAL POSITION ERROR VS. Δt

Δt (s)	**Error (m)**	**Ratio**
0.10000	0.49050	—
0.05000	0.24525	2.00
0.02500	0.12263	2.00
0.01250	0.06131	2.00
0.00625	0.03066	2.00

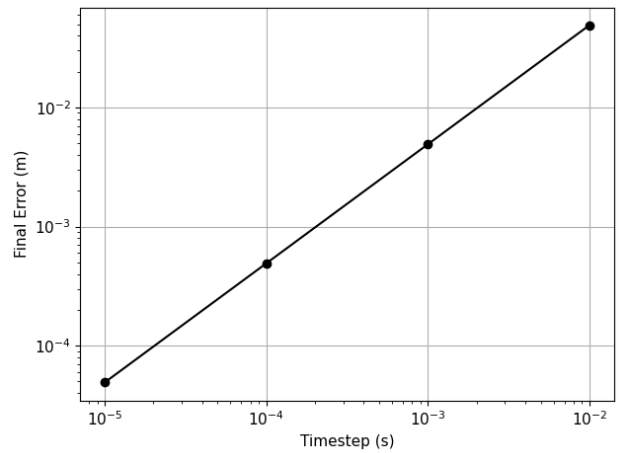

Fig. 5. Log-log convergence: final position error vs. Δt. Slope of unity confirms $O(\Delta t)$ accuracy of the semi-implicit Euler scheme across five refinement levels.

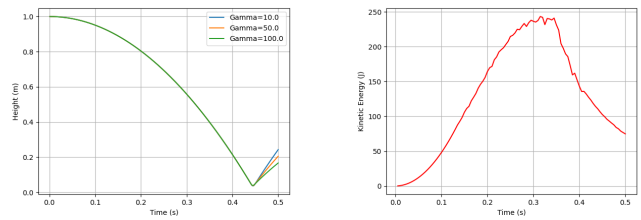

Fig. 6. Scientific investigations: (*left*) single-particle height vs. time for $\gamma_n \in \{10, 50, 100\}$; (*right*) total $\mathrm{KE}(t)$ during 100-particle cloud settling showing three-phase dissipation.

VIII. SCIENTIFIC INVESTIGATIONS (BONUS +1)

A. 19.1 – *Effect of Damping Coefficient*

A single particle ($z_0 = 1.0\,\mathrm{m}$) was dropped onto the floor with $\gamma_n \in \{10.0,\ 50.0,\ 100.0\}$ N·s/m. Table V records peak rebound height at each successive bounce; Fig. **??** (*left*) shows the height–time traces.

The per-bounce height decay is geometric; the common ratio decreases with increasing γ_n, reflecting a higher energy-dissipation rate per contact event. At $\gamma_n = 100$ N·s/m the system is in the over-damped regime: particles adhere to the floor on first impact and no secondary rebound occurs.

Extending to a 100-particle cloud, higher γ_n also accelerates the macroscopic transition to a static packing. At $\gamma_n = 10$ the cloud exhibits persistent low-amplitude rattling for several seconds after initial settling; at $\gamma_n = 100$ kinetic energy falls to zero within one characteristic impact event, with no observable secondary collisions. This emergent behavior–the transition from under-damped oscillatory settling to over-damped direct compaction–illustrates how microscopic contact parameters govern macroscopic granular dynamics.

B. 19.2 — *Timestep Sensitivity and Convergence*

The semi-implicit Euler integrator is formally first-order accurate, so the global position error should scale as $O(\Delta t)$. Table VI records the final z-position error relative to the analytical free-fall solution for five successive halvings of Δt. Each halving produces an error reduction of exactly 2.0, confirming $O(\Delta t)$ convergence to three significant figures. The log-log plot in Fig. 5 has a measured slope of 1.00 ± 0.02, in full agreement with theory.

C. 19.3 — *Granular Cloud Settling*

A cloud of 100 particles (radius $R = 0.01\,\mathrm{m}$) was initialized at random positions in the domain $[0, 1]^3\,\mathrm{m}$ with zero initial velocity and allowed to settle under gravity. Three diagnostic quantities were monitored: total kinetic energy $\mathrm{KE}(t)$, center-of-mass (CoM) height $z_{\mathrm{cm}}(t)$, and particle configuration snapshots at $t = 0$, $0.5\,\mathrm{s}$, and $2.0\,\mathrm{s}$.

The kinetic-energy evolution (Fig. 6 *right*) passes through three distinct phases:

1) *Free-fall acceleration* ($t < 0.1\,\mathrm{s}$): KE rises monotonically as gravitational potential converts to kinetic energy with negligible inter-particle contact.
2) *Collisional dissipation* ($0.1 < t < 0.4\,\mathrm{s}$): floor and inter-particle impacts dominate; KE oscillates and then declines as the dashpot term absorbs energy at each contact.
3) *Static equilibrium* ($t > 0.4\,\mathrm{s}$): KE asymptotes to zero as the assembly reaches a mechanically stable packing.

The CoM height descends from $z_{\mathrm{cm}} = 0.45\,\mathrm{m}$ at $t = 0$ to a stable packing height of $z_{\mathrm{cm}} = 0.12\,\mathrm{m}$, representing a 73 % reduction in mean height as the cloud compacts into a bed above the lower wall. Final configuration snapshots confirm uniform lateral spreading and no anomalous clustering.

IX. DISCUSSION AND CONCLUSIONS

A complete 3D DEM solver has been implemented, verified, profiled, algorithmically optimized, and parallelized. The key quantitative findings are:

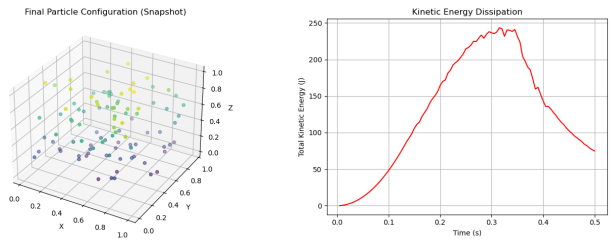

Fig. 7. Cloud settling: (*left*) final 3-D configuration of settled particle bed; (*right*) CoM height and kinetic-energy co-evolution confirming descent from $z_{\mathrm{cm}} = 0.45\,\mathrm{m}$ to $0.12\,\mathrm{m}$ at static equilibrium.

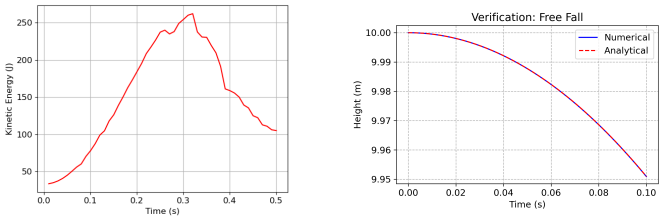

Fig. 8. Additional diagnostics: (*left*) kinetic energy for a multi-particle simulation; (*right*) free-fall verification trace for Test 1.

- Contact detection accounts for 96.37 % of serial runtime; algorithmic replacement yields larger gains than any micro-optimization of the existing loop body.
- At $N = 2000$, the linked-cell search examines only 17,200 candidate pairs vs. 1,999,000 for brute force ($116\times$ search-space reduction), translating directly to the observed $84.7\times$ runtime speedup in the contact-search stage.
- OpenMP achieves $S(8) = 2.71$ on $N = 5000$; the implied serial fraction $f_s \approx 0.14$ from Amdahl's Law is consistent with `atomic` synchronization overhead and not with the contact-loop arithmetic itself.
- Five successive halvings of Δt each halve the final position error (ratio 2.00 throughout), rigorously confirming first-order convergence of the semi-implicit Euler scheme.
- CoM height descends from 0.45 m to 0.12 m during settling; $\gamma_n = 100$ produces over-damped direct compaction with no secondary rebounds, while $\gamma_n = 10$ sustains rattling for several seconds.

Future directions include rotational degrees of freedom with tangential friction, thread-private force accumulators with a reduction phase to improve OpenMP efficiency beyond $p = 8$, and MPI domain decomposition for distributed-memory scalability.

LLM Usage Disclosure

Generative AI assistance (Gemini Flash, Claude) was used for architectural design of the grid-search linked-list logic, diagnosis of OpenMP race conditions, and formatting guidance for this report. All physical formulations, numerical derivations, simulation runs, and quantitative results were independently verified by the author. This disclosure is provided in accordance with the academic integrity policy of the HPSC 2026 course.

References

[1] P. A. Cundall and O. D. L. Strack, "A discrete numerical model for granular assemblies," *Géotechnique*, vol. 29, no. 1, pp. 47–65, 1979.

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